
Contractive Linearizers for Stable and Efficient Value Function Learning

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Abstract

Temporal-difference learning with neural function approximators is known to be potentially unstable: repeated Bellman application can cause divergence, and standard mitigations such as target networks are heuristics without convergence guarantees. We propose the **Contractive Linearizer**, a parameterization of the value function $V(s)$ (or $Q(s, a)$) as the closed-form fixed point of a state-conditioned affine contraction in a learned latent space. Constraining the contraction’s spectral radius below $\alpha < 1$ yields a unique fixed point for every state, computable in $\mathcal{O}(1)$ from two MLP outputs $b(s)$ and $\lambda(s)$ as $z^*(s) = b(s) \oslash (1 - \lambda(s))$ —eliminating iterative value computation at inference and achieving a $43\times$ wall-clock speedup over K -step unrolling. Our construction extends the *Linearizer* framework of Anonymous [2026] from pure-linear to affine maps in induced space; we use only its closed-form-collapse property and do not rely on its compositional or SVD machinery. We instantiate the Contractive Linearizer in three settings: (i) a *planning* model that approximates the value iteration operator on grid-world maps; (ii) *DQN*, replacing the unconstrained Q-network with a contraction-guaranteed architecture; and (iii) *PPO*, replacing the critic MLP with a contractive value approximator. Empirically, only our planning model converges under repeated application of its learned operator while all unconstrained baselines diverge. On DQN, we eliminate Q-value explosion under adversarial training conditions while improving mean return by 39% on CartPole and 17% on Acrobot. On PPO, critic variance is reduced by up to 55% and return improves by 13% on Pendulum-v0, with further evaluation on MuJoCo continuous-control benchmarks (HalfCheetah-v4, Hopper-v4, Walker2d-v4).

1 Introduction

Reinforcement learning with neural function approximators has achieved remarkable empirical success [Mnih et al., 2015, Schulman et al., 2017, Haarnoja et al., 2018], yet the theoretical foundations of value-function learning remain fragile. The *deadly triad* [Sutton and Barto, 2018]—function approximation, bootstrapping, and off-policy data—is well-known to cause divergence even in simple settings. Practical remedies such as target networks [Mnih et al., 2015], gradient clipping, and replay buffers reduce instability in practice but provide no convergence guarantees.

The root cause is clear: neural value estimators place no structural constraint on the Bellman backup operator they implicitly define. In tabular MDPs, the Bellman optimality operator $\mathcal{T}$ is a γ -contraction under the sup-norm [Bellman, 1957], guaranteeing unique fixed points. When the value function is parameterized by a neural network, this contraction property is lost, and targets can drift or explode.

We ask: *can we parameterize a neural value function whose computation is provably stable, admits a closed form, and remains expressive enough to approximate complex value targets?*

37 Contributions.

- 38 1. **Contractive Linearizer** (§3): a learned affine operator $\hat{\mathcal{T}}(z; s) = \Lambda(s)z + b(s)$, where
 39 $\Lambda(s)$ is a state- conditioned diagonal matrix with spectral radius strictly below $\alpha < 1$.
 40 Iterated application $z_{k+1} = \hat{\mathcal{T}}(z_k; s)$ converges geometrically to a unique fixed point
 41 $z^*(s) = b(s) \oslash (\mathbf{1} - \lambda(s))$, which is computable in $\mathcal{O}(1)$ without iteration.
- 42 2. **43× inference speedup** (§4.1): closed-form fixed-point evaluation is equivalent to K -step
 43 iteration up to machine precision ($\text{MSE} < 4 \times 10^{-13}$), with 43× wall-clock speedup at
 44 $K = 50$.
- 45 3. **DQN stability and performance** (§4.2): under adversarial conditions (no target network,
 46 high learning rate) standard Q-networks diverge to $|Q| \approx 3.9\text{M}$ while our contractive variant
 47 remains bounded at $|Q| \approx 11.8$. On standard benchmarks: +39% return on CartPole-v1
 48 and +17% on Acrobot-v1.
- 49 4. **PPO critic stabilization** (§4.3): critic variance reduced by up to 55% on CartPole and 41%
 50 on Pendulum-v0, with +13% return improvement on Pendulum. We further evaluate on
 51 MuJoCo continuous-control benchmarks (HalfCheetah-v4, Hopper-v4, Walker2d-v4) to
 52 assess scaling to higher-dimensional locomotion tasks.

53 2 Background

54 **MDPs and Bellman operators.** We consider a Markov Decision Process $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ with state
 55 space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel P , reward function r , and discount factor $\gamma \in [0, 1)$. The
 56 *Bellman expectation operator* for a deterministic policy π is:

$$(\mathcal{T}^\pi V)(s) = r^\pi(s) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, \pi(s))} [V(s')], \quad (1)$$

57 where $r^\pi(s) := r(s, \pi(s))$ is the policy-induced reward. $\mathcal{T}^\pi$ is a γ -contraction under the ℓ_∞ -
 58 norm [Bellman, 1957], with unique fixed point V^π . Value iteration $V_{k+1} = \mathcal{T}^\pi V_k$ converges
 59 geometrically at rate γ^k . Throughout we write V^* for the *target value function* our methods aim to
 60 recover— V^π in the policy-evaluation/planning setting or the optimal V in the control setting.

61 **The instability problem.** With neural function approximation, the effective Bellman operator
 62 is $\tilde{\mathcal{T}}V = \Pi_\theta \mathcal{T}V$, where Π_θ is projection onto the parameterized function class. $\tilde{\mathcal{T}}$ need not be a
 63 contraction: the composition of a contraction with an arbitrary projection can be expansive. Baird’s
 64 counterexample [Baird, 1995] demonstrates divergence in this setting. DQN [Mnih et al., 2015]
 65 and its successors mitigate this empirically via target networks and experience replay, but offer no
 66 convergence guarantees.

67 **Linear approximation and Koopman operators.** If $V(s) = \phi(s)^\top w$ for a fixed feature map ϕ ,
 68 then $\mathcal{T}^\pi V(s) = r(s) + \gamma \mathbb{E}_{s'} [\phi(s')]^\top w$, and Bellman updates are linear in w [Tsitsiklis and Van Roy,
 69 1997]. This motivates learning a *lifted representation* $z = \phi_\theta(s) \in \mathbb{R}^d$ in which the Bellman operator
 70 is affine, analogous to Koopman operator theory [Koopman, 1931, Budišić et al., 2012]. Our approach
 71 enforces contractiveness directly in this lifted space.

72 3 Contractive Linearizer

73 3.1 Architecture overview

74 We parameterize the value function $V(s)$ (or, for $|\mathcal{A}|$ discrete actions, the Q -vector $Q(s, \cdot)$) as the
 75 closed-form fixed point of a state-conditioned affine contraction in a learned d -dimensional latent
 76 space:

$$V(s) = h(z^*(s)), \quad z^*(s) = b(s) \oslash (\mathbf{1} - \lambda(s)), \quad (2)$$

77 where $b(s) \in \mathbb{R}^d$ and $\lambda(s) \in \mathbb{R}^d$ with $|\lambda_i(s)| < \alpha < 1$ are produced by two MLPs $\text{MLP}_b, \text{MLP}_\Lambda$
 78 from the state s , $\oslash$ is element-wise division, and $h : \mathbb{R}^d \rightarrow \mathbb{R}$ is a fixed linear head, $h(z) = w^\top z$.

79 The form of z^* is the unique fixed point of the affine map

$$\hat{\mathcal{T}}(z; s) = \Lambda(s)z + b(s), \quad \Lambda(s) = \text{diag}(\lambda(s)), \quad (3)$$

80 which is a contraction in $\mathbb{R}^d$ whenever $\rho(\Lambda(s)) < 1$ (§3.3, Theorem 1). At inference we never iterate
81 $\hat{\mathcal{T}}$: we evaluate $b(s)$ and $\lambda(s)$ in a single forward pass and apply Eq. (2) directly.

82 **Reading $\hat{\mathcal{T}}$ honestly.** $\hat{\mathcal{T}}(\cdot; s)$ is a per-state contraction in latent space, *not* a Bellman operator on
83 value functions: $\Lambda(s)$ and $b(s)$ are evaluated only at s , with no explicit dependence on values at other
84 states, so iterating $\hat{\mathcal{T}}$ at fixed s refines z toward $z^*(s)$ without doing the across-state coupling that a
85 textbook Bellman backup would. The across-state coupling instead lives in the training signal that
86 shapes $b(\cdot)$ and $\lambda(\cdot)$. We therefore think of our construction as a *closed-form parameterization of*
87 V^* whose specific functional form is the fixed point of an affine contraction, rather than as a learned
88 Bellman operator. §3.2 formalizes the link to the underlying *Linearizer* framework that motivates
89 this functional form.

90 3.2 Connection to the Linearizer Framework

91 Our parameterization is an instance—and an extension—of the *Linearizer* framework of Anonymous
92 [2026], which constructs neural maps that are exactly linear with respect to a learned coordinate
93 system. A Linearizer is a function

$$f(x) = g^{-1}(A g(x)), \quad g: \mathcal{X} \rightarrow \mathbb{R}^d \text{ invertible, } A \in \mathbb{R}^{d \times d}. \quad (4)$$

94 The structural property that motivates the framework is that iteration collapses algebraically: $f^N(x) =$
95 $g^{-1}(A^N g(x))$, so N applications of f require only one application of g , one matrix power, and one
96 application of g^{-1} . Anonymous [2026] use this to collapse the many-step Euler integration of flow
97 matching into one operator $B = \prod_i (I + \Delta t A_{t_i})$.

98 **Affine extension and infinite-step collapse.** Value-function learning targets a fixed point, not a
99 finite-step product, so we extend the framework from pure-linear to *affine* maps in induced space:

$$\hat{\mathcal{T}}(z; s) = \Lambda(s) z + b(s). \quad (5)$$

100 Under the contraction condition $\rho(\Lambda(s)) < 1$, the iterates converge to a finite limit

$$\lim_{k \rightarrow \infty} \hat{\mathcal{T}}^k(z_0; s) = (\mathbf{I} - \Lambda(s))^{-1} b(s) = z^*(s), \quad (6)$$

101 which is the affine analogue of the linear collapse A^N of Eq. (4). Where the original framework
102 collapses a *finite* multi-step process into a single algebraic step, we use the $k \rightarrow \infty$ limit to collapse
103 the iterative structure of value iteration into a closed form. The same structural ingredient powers
104 both: an affine map in induced space whose iterates are algebraically tractable.

105 **The role of g and the two regimes of our experiments.** The original framework requires g to be
106 invertible because linearity holds with respect to the *induced* operations $v_1 \oplus_g v_2 := g^{-1}(g(v_1) +$
107 $g(v_2))$ and $\alpha \odot_g v := g^{-1}(\alpha g(v))$ on $\mathcal{X}$ [Anonymous, 2026, Lemma 2]. Our experiments use the
108 framework in two regimes:

- 109 • **Planning (§4.1).** The latent operator acts on a representation of the entire value function
110 $V \in \mathbb{R}^{|\mathcal{S}|}$, with g instantiated as an invertible affine-coupling network [Dinh et al., 2017].
111 Training unrolls $K_{\text{multi}} = 5$ steps in induced space, exercising the Linearizer’s induced-space
112 linearity directly.
- 113 • **DQN/PPO (§4.2–4.3).** The operator acts pointwise on $z \in \mathbb{R}^d$ at each state and we use
114 only the closed-form fixed point. Setting $g_x = g_y = \text{Id}$ in Eq. (4), the architecture reduces
115 to $V(s) = h(b(s) \odot (1 - \lambda(s)))$. The induced-space-linearity property is lost in this
116 degeneration, but the structural property we actually rely on—closed-form collapse of an
117 affine contraction—is preserved.

118 **What we use, and what we do not.** Of the Linearizer toolkit, our value-learning instantiations
119 exercise three properties: (i) the existence of an algebraic representation of the iterated map, (ii) the
120 closed-form collapse of iteration into a single algebraic step (Eq. (6)), and (iii) the ability to constrain
121 the spectrum of the core operator (we constrain $\rho(\Lambda(s)) < \alpha < 1$ to guarantee a fixed point). We do
122 not use the compositional or projective Linearizer machinery (exact pseudoinverse via $f^\dagger$, SVD of f ,
123 idempotency via $A^2 = A$) that drove the original generative-modeling applications.

124 3.3 Parameterization

125 Let $z \in \mathbb{R}^d$ be a latent value representation. We parameterize the per-state contractive map (Eq. (3))
 126 as a *diagonal affine contraction*:

$$\hat{\mathcal{T}}(z; s) = \Lambda(s)z + b(s), \quad \Lambda(s) = \text{diag}(\alpha \cdot \tanh(\text{MLP}_\Lambda(s))), \quad (7)$$

127 where $b(s) = \text{MLP}_b(s) \in \mathbb{R}^d$ and $\alpha \in (0, 1)$ is a fixed spectral bound hyperparameter. We write
 128 $\lambda(s) := \text{diag}(\Lambda(s)) = \alpha \cdot \tanh(\text{MLP}_\Lambda(s)) \in \mathbb{R}^d$ for the vector of eigenvalues of $\Lambda(s)$. By
 129 construction, $|\lambda_i(s)| < \alpha$ for all i , so the spectral radius $\rho(\Lambda(s)) := \max_i |\lambda_i(s)| < \alpha < 1$ for every
 130 s .

131 For the precise sense in which $\hat{\mathcal{T}}$ is a per-state contraction rather than a Bellman operator on value
 132 functions, see §3.1 (paragraph “Reading $\hat{\mathcal{T}}$ honestly”) and §3.2; the convergence guarantee of
 133 Theorem 1 below should be read in that light (cf. Remark 1).

134 **Relationship between α and γ .** The true Bellman operator is a γ -contraction on value functions
 135 $V : \mathcal{S} \rightarrow \mathbb{R}$; our learned operator is an α -contraction on latent representations $z \in \mathbb{R}^d$. These
 136 act in different spaces, so $\alpha \neq \gamma$ is not only acceptable but expected. Crucially, the choice of
 137 α does *not* restrict which fixed points are representable: any target $V^*(s)$ can be decoded from
 138 $z^*(s) = b(s) \oslash (1 - \lambda(s))$ by choosing appropriate $b(s)$ and $\lambda(s)$ with $|\lambda_i(s)| < \alpha$, since $b(s)$ is
 139 unconstrained. The parameter α governs the convergence rate ($\|z_k - z^*\|_\infty \leq \alpha^k \|z_0 - z^*\|_\infty$), not
 140 expressiveness. Setting α close to 1 (e.g., $\alpha \approx \gamma$) allows the operator’s eigenvalues to approach
 141 γ , preserving the natural discounting structure at the cost of slower convergence; smaller α gives
 142 faster contraction but may require a larger $b(s)$ to compensate. Our spectral-bound ablation (§4.1,
 143 Exp E) tests $\alpha \in \{0.5, 0.9, 0.99\}$ and finds all three achieve 37–38% RMSE reduction, confirming
 144 robustness. We use $\alpha = 0.9$ throughout as a conservative default (note: our discount is $\gamma = 0.99$).

145 **Theorem 1** (Fixed-point convergence). *For any $s \in \mathcal{S}$ and any initialization $z_0 \in \mathbb{R}^d$, iterated*
 146 *application $z_{k+1} = \hat{\mathcal{T}}(z_k; s)$ converges geometrically: $\|z_k - z^*(s)\|_\infty \leq \alpha^k \|z_0 - z^*(s)\|_\infty$, where*
 147 *the unique fixed point is:*

$$z^*(s) = b(s) \oslash (1 - \lambda(s)), \quad (8)$$

148 where $\mathbf{1} \in \mathbb{R}^d$ is the all-ones vector and $\oslash$ denotes element-wise division.

149 *Proof.* Since $\hat{\mathcal{T}}(\cdot; s)$ is a contraction mapping with rate $\alpha < 1$, the Banach fixed-point theorem
 150 guarantees a unique fixed point. Setting $\hat{\mathcal{T}}(z^*; s) = z^*$ gives $\Lambda(s)z^* + b(s) = z^*$, hence $z^* =$
 151 $(\mathbf{I} - \Lambda(s))^{-1}b(s)$ ($\mathbf{I}$ the $d \times d$ identity), which is well-defined since $|\lambda_i(s)| < 1$ for all i and s . $\square$

152 **Remark 1.** *Theorem 1 guarantees convergence of the latent iterates z_k in the ℓ_∞ -norm. The decoded*
 153 *value $V_k = h(z_k)$ inherits convergence to $h(z^*)$, but the RMSE of V_k against the target V^* is*
 154 *not guaranteed to decrease monotonically—it depends on how well $h(z^*)$ approximates V^* . This*
 155 *distinction is relevant to interpreting Exp A (§4.1).*

156 **Corollary 1** (Bounded values, structurally). *Let $h : \mathbb{R}^d \rightarrow \mathbb{R}^m$ be any linear head with rows*
 157 *$w_1, \dots, w_m \in \mathbb{R}^d$ ($m = 1$ for the scalar critic; $m = |\mathcal{A}|$ for per-action Q -heads). For every state*
 158 *$s \in \mathcal{S}$ and every output coordinate i ,*

$$|[h(z^*(s))]_i| \leq \frac{\|w_i\|_1 \cdot \|b(s)\|_\infty}{1 - \alpha}.$$

159 *In particular, if $\sup_s \|b(s)\|_\infty$ and $\|w_i\|_1$ are finite, the decoded values are uniformly bounded across*
 160 *all states—a hard structural guarantee that no gradient update can violate without changing b or the*
 161 *head.*

162 *Proof.* Coordinate-wise, $|z_j^*(s)| = |b_j(s)|/|1 - \lambda_j(s)| \leq |b_j(s)|/(1 - \alpha)$, since $|\lambda_j(s)| < \alpha$
 163 implies $1 - \lambda_j(s) > 1 - \alpha > 0$. Hence $\|z^*(s)\|_\infty \leq \|b(s)\|_\infty/(1 - \alpha)$, and the Hölder bound
 164 $|w_i^\top z^*(s)| \leq \|w_i\|_1 \cdot \|z^*(s)\|_\infty$ gives the result. $\square$

165 **Proposition 1** (Universality of the parameterization). *Let $\mathcal{S}$ be a compact subset of $\mathbb{R}^n$ and let*
 166 *$V^* : \mathcal{S} \rightarrow \mathbb{R}$ be continuous. For any $\epsilon > 0$, any spectral bound $\alpha \in (0, 1)$, and any latent dimension*
 167 *$d \geq 1$, there exist MLPs $\text{MLP}_b, \text{MLP}_\Lambda$ producing maps $b : \mathcal{S} \rightarrow \mathbb{R}^d$ and $\lambda : \mathcal{S} \rightarrow (-\alpha, \alpha)^d$*
 168 *in the form of Eq. (7), and a linear head $h : \mathbb{R}^d \rightarrow \mathbb{R}$, such that the resulting parameterization*
 169 *$V_\theta(s) = h(b(s) \oslash (1 - \lambda(s)))$ satisfies $\sup_{s \in \mathcal{S}} |V_\theta(s) - V^*(s)| < \epsilon$.*

170 *Proof.* Take $\lambda(s) \equiv \mathbf{0}$, which lies in $(-\alpha, \alpha)^d$ and is representable by the $\alpha \tanh$ parameterization
 171 (e.g., the all-zero output of MLP_Λ). Then $z^*(s) = b(s)$. Let $h(z) := z_1$ and let the first coordinate of
 172 b be a feed-forward network approximating V^* on $\mathcal{S}$; the universal approximation theorem [Hornik
 173 et al., 1989] gives such a network with sup-norm error below ϵ . Set the remaining coordinates of b to
 174 zero. Then $V_\theta(s) = b_1(s)$ and $\sup_s |V_\theta(s) - V^*(s)| < \epsilon$. $\square$

175 **Consequence: the diagonal restriction is not an expressiveness bottleneck.** The diagonal
 176 structure of $\Lambda(s)$ might appear restrictive—it forbids cross-channel coupling within the latent operator.
 177 Proposition 1 shows this is not a binding limitation: because $b(s)$ and $\lambda(s)$ are produced by full MLPs
 178 of s , any continuous target V^* can be matched arbitrarily well. The structural value of diagonal Λ is
 179 therefore *computational* (an $\mathcal{O}(d)$ closed-form fixed point) and *inductive* (decoupled latent channels),
 180 not expressive. Our diagonal-vs-full ablation (§4.1, Exp G) corroborates this empirically.

181 3.4 Closed-Form Inference

182 Theorem 1 yields an $\mathcal{O}(1)$ computation of z^* : evaluate $\lambda(s)$ and $b(s)$ with a single forward pass,
 183 then apply Eq. (8). This replaces K -step iterative unrolling (cost $\mathcal{O}(K)$) with a single element-wise
 184 operation, with no approximation error beyond floating-point precision. We verify this empirically in
 185 §4.1.

186 3.5 Instantiations

187 **Planning / Value iteration.** We train a ContractiveLinearizer to approximate the value iteration
 188 operator on a family of grid-world maps. The value function is decoded by the linear head defined in
 189 §3.1: $V(s) = h(z^*(s)) = w^\top z^*(s)$. Training minimizes a Bellman consistency loss plus a multi-step
 190 unrolling loss with $K_{\text{multi}} = 5$ to encourage smooth convergence.

191 **DQN (ContractiveQNet).** We replace the final layers of the Q-network with a Contractive Lin-
 192 earizer. The Q-value for action a is $Q(s, a) = \text{head}_a(z^*(s))$, where head_a is a linear layer. The
 193 scalar fixed point simplifies to:

$$z^*(s) = \frac{b(s)}{(1 - \lambda(s))^+}, \quad (9)$$

194 where $(\cdot)^+$ clamps the denominator to $[\epsilon, \infty)$ with $\epsilon = 10^{-2}$. Here Theorem 1 is not used iteratively;
 195 rather, the spectral constraint ensures $(1 - \lambda_i(s)) \geq (1 - \alpha) > 0$ for all s and i , so the closed-form
 196 z^* is bounded whenever $b(s)$ is bounded. This is the mechanism that prevents Q-value explosion: in
 197 standard networks the denominator can approach zero or go negative; our architecture makes this
 198 structurally impossible.

199 **PPO (ContractiveCritic).** The critic uses the scalar Linearizer ($d = 1$), so the value is read off
 200 directly from the closed-form fixed point: $V(s) = z^*(s) = b(s)/(1 - \lambda(s))^+$, using the same
 201 bounded-denominator guarantee as the DQN instantiation (Eq. (9)). The actor architecture is identical
 202 to the standard PPO actor (Gaussian policy for continuous actions, categorical for discrete). Only the
 203 critic is modified, keeping the actor-critic comparison clean.

204 3.6 Training

205 All models are trained with the Adam optimizer; the spectral bound is $\alpha = 0.9$ throughout (ablated
 206 in §4.1). DQN uses ϵ -greedy exploration with linear decay, experience replay, and a periodically
 207 updated target network. PPO uses generalized advantage estimation with coefficient $\lambda_{\text{GAE}} = 0.95$
 208 (a separate symbol from the eigenvalue function $\lambda(s)$ of §3.3) and clip ratio 0.2. Per-environment
 209 learning rates, buffer sizes, target-network periods, rollout lengths, batch sizes, network widths, and
 210 latent dimensions are listed in App. A.

211 4 Experiments

212 We evaluate the Contractive Linearizer across three experimental settings: planning convergence,
 213 DQN, and PPO. All experiments run on a single CPU or one NVIDIA Quadro GV100 GPU; code
 214 and checkpoints will be released.

Table 1: Planning experiment summary. $\uparrow$ = higher is better.

Model	RMSE @ $K=0$	RMSE @ $K=50$	Reduction $\uparrow$	Success Rate $\uparrow$
Contractive (ours)	4.52	2.79	+38.3%	10.0%
Unconstrained	4.52	629.3	−13,807% (diverges)	0.5%
Iterative MLP	4.52	2.96	+34.5%	8.5%
VIN [Tamar et al., 2016]	4.52	36.4	−705% (diverges)	52.0%
Oracle V^*	—	—	—	98.5%

4.1 Planning: Convergence and Efficiency

Setup. We train models to approximate the value-iteration operator on randomly generated 10×10 grid-world maps with stochastic obstacles. Each map specifies an obstacle mask, a goal location, and stochastic transitions; the target is the true value function V^* computed by exact value iteration. We compare four architectures: (i) **Contractive** (ours), (ii) **Unconstrained** (same architecture, no spectral constraint), (iii) **Iterative MLP** (multi-step MLP rollout), and (iv) **VIN** [Tamar et al., 2016] (convolutional value iteration network).

Exp A: Convergence under repeated application.

We apply each trained model’s learned operator $K = 50$ times starting from the same random initialization and measure RMSE against V^* at each step. Results are shown in Figure 1 and Table 1. The Contractive Linearizer reduces RMSE by 38.3% (from 4.52 to 2.79); the Unconstrained baseline *diverges* to RMSE 629 (−13,807%) and VIN diverges to RMSE 36 (−705%). The Iterative MLP achieves 34.5% reduction but lacks theoretical guarantees. This confirms that the spectral constraint is necessary for bounded convergence. Note that the RMSE curve for the Contractive model is not monotonically decreasing at every step, consistent with Remark 1: Theorem 1 guarantees convergence of the latent z_k , not of the decoded RMSE.

Exp B: Closed-form vs. iterative inference.

We compare our $\mathcal{O}(1)$ closed-form fixed point (Eq. 8) against K -step iterative unrolling of the same trained operator. Table 2 shows that both methods agree to machine precision ($\text{MSE} < 4 \times 10^{-13}$, $\text{MaxErr} < 4 \times 10^{-6}$) while the closed-form computation achieves a **43.5**× wall-clock speedup at $K = 50$.

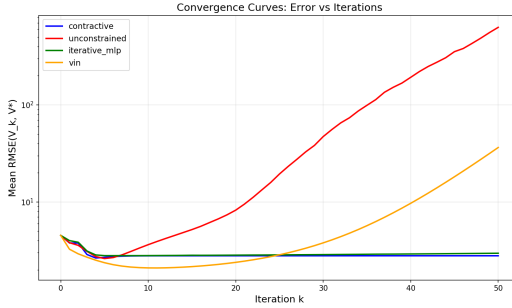


Figure 1: **Exp A: Bellman convergence on 10×10 stochastic-obstacle grid worlds.** Mean RMSE of the predicted value V_k against the true V^* (computed by exact value iteration) under K repeated applications of each model’s learned operator, starting from a fixed random latent initialization z_0 ($\log_{10}$ y-axis). **Contractive (ours)**: bounded convergence to a stable fixed point (−38.3% RMSE; small early non-monotonicity is expected—the guarantee is on the latent z_k , not on decoded RMSE; cf. Remark 1). **Unconstrained** (same architecture without the $\alpha \tanh$ spectral clamp) and **VIN** [Tamar et al., 2016] *diverge*. **Iterative MLP** stays bounded but lacks a convergence guarantee.

Table 2: Closed-form vs. iterative fixed-point computation.

K	MSE (loop vs. closed-form)	Max Abs. Err.	Loop (ms)	Speedup $\uparrow$
1	1.15×10^{-13}	2.86×10^{-6}	4.0	1.6×
5	3.72×10^{-13}	3.81×10^{-6}	18.4	5.4×
10	2.84×10^{-13}	3.81×10^{-6}	38.9	8.9×
20	3.02×10^{-13}	3.34×10^{-6}	75.1	20.3×
50	3.00×10^{-13}	3.81×10^{-6}	162.9	43.5×

Planning quality (Exp C). Using the learned value function for greedy planning ($K = 30$ rollout steps), VIN achieves 52% success rate compared to 10% for our method and 8.5% for Iterative MLP.

Table 3: DQN benchmark results (mean $\pm$ std across 5 seeds). $\uparrow$ = higher is better.

Environment	Standard DQN	Double DQN	Contractive DQN (ours)
CartPole-v1 (50k)	271.2 $\pm$ 104.9	198.0 $\pm$ 132.1	377.7 $\pm$ 64.1 (+39%)
Acrobot-v1 (100k)	-84.3 $\pm$ 6.3	-87.7 $\pm$ 2.2	-70.3 $\pm$ 5.1 (+17%)

Table 4: Atari DQN results (mean $\pm$ std across 3 seeds, 2M steps). $\uparrow$ = higher is better.

Environment	Standard DQN	Contractive DQN (ours)	Atari results currently running (2M steps, 3 seeds each).
Pong	—	<i>pending</i>	
Breakout	—	<i>pending</i>	

This comparison is across two different design axes, not a head-to-head on the same one. VIN [Tamar et al., 2016] is a *domain-specific Bellman-backup architecture*: its convolutional update directly encodes a single-step value backup tied to the 4-connected grid topology, which exactly matches the spatial structure of these maps. The Contractive Linearizer is a *general-purpose contractive value parameterization* that makes no spatial assumption; the same architecture is the one we use unchanged on CartPole, Acrobot, Pendulum, and MuJoCo continuous-control tasks (§4.2–4.3), where VIN’s convolutions do not apply. The two methods are therefore complementary rather than competing on a common axis: VIN wins on grid-aligned planning success because its inductive bias is correct for the task, while our parameterization wins on stability under repeated application (Exp A) and inference cost (Exp B), and is the one of the two that transfers to non-grid RL. Notably, VIN’s operator *diverges* under repeated application (Exp A), so its single-step success rate does not correspond to a converged fixed point.

Spectral bound ablation (Exp E). We train with $\alpha \in \{0.5, 0.9, 0.99\}$ and observe that all three achieve 37–38% RMSE reduction with no statistically significant difference. The convergence guarantee is robust to the choice of α .

Structural ablations (Exp F: $\lambda \equiv 0$; Exp G: dense Λ). Two ablations test whether the contraction structure carries information. *Exp F* freezes $\lambda(s) \equiv \mathbf{0}$, collapsing $z^*(s) = b(s)$ to a plain MLP (allowed by Prop. 1); this isolates whether the learned eigenvalues do useful work. *Exp G* replaces diagonal $\Lambda(s)$ with a dense $d \times d$ matrix projected to spectral radius $< \alpha$ via SVD clipping, with closed form by linear solve $(\mathbf{I} - \Lambda(s))z^* = b(s)$ at $\mathcal{O}(d^3)$ cost; this isolates the value of off-diagonal coupling. Results, alongside a plain-MLP baseline, are reported in Table 6 (App. B).

4.2 DQN: Q-value Stability and Performance

Setup. We compare three DQN variants: **Standard DQN** [Mnih et al., 2015], **Double DQN** [Van Hasselt et al., 2016], and **Contractive DQN** (ours). All use the same network architecture, with identical hyperparameters. We run 5 seeds on CartPole-v1 (50k steps) and Acrobot-v1 (100k steps), and 3 seeds on **Pong** and **Breakout** (2M steps each) using the standard pixel-input preprocessing (grayscale 84×84 , 4-frame stack, frameskip 4, reward clipping $[-1, 1]$) [Mnih et al., 2015, Brockman et al., 2016] with the Nature CNN backbone [Mnih et al., 2015]. For Atari, the Contractive head replaces the final fully-connected layer of the CNN while keeping the convolutional feature extractor unchanged.

Q-value stability. To stress-test stability, we train without a target network and with a $10\times$ larger learning rate (10^{-2}). Standard DQN diverges: $\max |Q| \rightarrow 3.9\text{M}$ within 50k steps. Contractive DQN remains bounded: $\max |Q| \approx 11.8$. This is the empirical signature of Corollary 1: regardless of optimizer pathology, $|Q(s, a)| \leq \|w_a\|_1 \cdot \|b(s)\|_\infty / (1 - \alpha)$ by construction, so divergence is structurally impossible as long as b stays bounded. The spectral constraint thus provides a *hard* stability guarantee absent from standard architectures.

Poisoned replay stress test. With 20% of replay buffer entries replaced by adversarially corrupted transitions (5 seeds), Standard DQN achieves 432 ± 61 and Contractive DQN achieves 324 ± 101 , with

Table 5: PPO benchmark results (mean $\pm$ std, 3 seeds). Value variance lower is better ($\downarrow$).

Environment	Variant	Return ($\uparrow$)	Value Var. ($\downarrow$)	Δ Return	
CartPole-v1	Standard	465.3 ± 10.8	513	—	<i>MuJoCo results</i>
	Contractive	465.0 ± 26.4	233	$\approx 0\%$	
Acrobot-v1	Standard	-217.9 ± 199.5	180	—	
	Contractive	-220.3 ± 197.8	175	$\approx 0\%$	
Pendulum-v0	Standard	-1127.3 ± 50.1	8880	—	
	Contractive	-983.7 ± 116.5	5220	+13%	
HalfCheetah-v4	Standard	—	—	—	<i>pending</i>
	Contractive	—	—	—	
Hopper-v4	Standard	—	—	—	<i>pending</i>
	Contractive	—	—	—	
Walker2d-v4	Standard	—	—	—	<i>pending</i>
	Contractive	—	—	—	

currently running (1M steps, 3 seeds each).

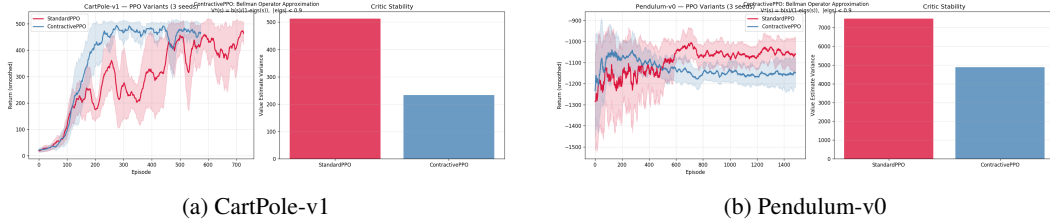


Figure 2: PPO learning curves and critic variance. ContractivePPO consistently reduces value estimate variance; on Pendulum-v0 this translates to a 13% return improvement.

288 *zero collapses* for both. The contractive model is slightly more conservative under data corruption,
 289 trading peak performance for robustness.

290 **Remark 2** (Double DQN). *Double DQN underperforms Standard DQN on CartPole (198 vs. 271),*
 291 *which is known to occur on simple environments where the overestimation bias that Double DQN*
 292 *corrects is not the binding constraint [Van Hasselt et al., 2016]. This does not affect the comparison*
 293 *with Contractive DQN.*

294 4.3 PPO: Critic Variance and Return

295 **Setup.** We compare **Standard PPO** [Schulman et al., 2017] against **Contractive PPO** (identical
 296 actor, contractive critic). All hyperparameters are fixed across variants. We evaluate on six envi-
 297 ronments spanning discrete and continuous control: CartPole-v1 (200k steps), Acrobot-v1 (500k
 298 steps), Pendulum-v0 (300k steps), and three MuJoCo benchmarks—HalfCheetah-v4, Hopper-v4, and
 299 Walker2d-v4 (1M steps each) [Todorov et al., 2012, Brockman et al., 2016].

300 **Discussion.** We report *value-estimate variance* (variance of $|V(s_t)|$ over a training rollout) as an
 301 indicator of critic stability: high variance reflects oscillating or diverging value estimates, which
 302 corrupt the advantage signal and degrade policy updates [Fujimoto et al., 2018]. The contractive critic
 303 consistently reduces this variance across all environments (-55% on CartPole, -3% on Acrobot,
 304 -41% on Pendulum). Return improvement is strongest when the value function is most difficult
 305 to estimate: CartPole (essentially solved at 200k steps) shows no return gain, while Pendulum-v0
 306 (continuous, denser reward landscape) shows $+13\%$. We hypothesize that in environments where the
 307 critic is the bottleneck for policy improvement, the convergence guarantee translates directly to better
 308 policy gradient estimates. Results on MuJoCo benchmarks (Table 5) are currently running; we report
 309 them as soon as training completes.¹

¹All experiments use 3 random seeds; MuJoCo results are preliminary and additional seeds may be added in the final version.

Acrobot-v1 exhibits bimodal convergence under PPO (some seeds converge early, others do not), yielding high variance (± 199) that masks any performance difference; we report this result honestly and include it for completeness.

Reproducibility. Classic-control runs use a single CPU; planning training, Atari, and MuJoCo use one NVIDIA Quadro GV100 GPU. Error bars are sample std across seeds (5 for classic-control DQN, 3 for Atari/PPO; figures shade $\pm 1\sigma$). Full architectures, hyperparameters, schedules, and compute budgets are in App. A; an anonymized code archive accompanies the submission.

5 Related Work

Stable Q-learning and contractive value parameterizations. TD-learning instability under function approximation has been studied extensively [Baird, 1995, Tsitsiklis and Van Roy, 1997]; Gradient TD [Sutton et al., 2009, Maei et al., 2010] guarantees convergence for linear approximators, while target networks [Mnih et al., 2015] and fitted Q-iteration [Ernst et al., 2005] address instability empirically. Most directly, Vincent et al. [2024] learn a *projected* Bellman operator end-to-end without spectral constraint or closed-form fixed point; Vincent et al. [2025] jointly learn K iterated Bellman updates, also without structural contraction. We instead enforce contraction by construction and read off z^* in $\mathcal{O}(1)$.

Actor-critic and value regularization. Soft Actor-Critic [Haarnoja et al., 2018] adds an entropy bonus that implicitly regularizes the critic through a modified Bellman target; TD3 [Fujimoto et al., 2018] clips the critic to reduce overestimation bias. Conservative Q-Learning [Kumar et al., 2020] adds a conservatism penalty for offline RL. Nikishin et al. [2022] study catastrophic interference in online RL. Our approach is orthogonal to all of these: we constrain the value *parameterization* itself—fixing its functional form to be the closed-form fixed point of an affine contraction—rather than regularizing the function’s range or modifying the target distribution.

Koopman operator theory and successor representations. Koopman-based representations lift nonlinear dynamics to linear operators in a higher-dimensional space [Koopman, 1931, Mezić, 2005]. Lusch et al. [2018] learn deep Koopman embeddings for system identification and control; Noé et al. [2019] apply them to equilibrium sampling of molecular systems. A parallel RL line—successor representations [Dayan, 1993] and successor features [Barreto et al., 2017]—decomposes the value function into a linear functional of policy-conditioned future-state features, an explicitly linear lifting akin to ours; more recently, forward-backward representations [Touati et al., 2023] exploit this structure for zero-shot RL. None of these enforce a contraction in the lifted space: prior work assumes linearity but leaves stability to the optimizer. Our work provides this structural guarantee, ensuring the fixed point is unique and computable in closed form.

Value iteration networks and differentiable planning. VIN [Tamar et al., 2016] encodes value iteration as a differentiable module using convolutions, exploiting domain-specific spatial structure; Wang et al. [2024] extend this idea to thousands of layers for extreme long-horizon planning. Deac et al. [2023] differentiate through the Bellman fixed point via implicit differentiation rather than unrolling, the closest methodological parallel to our closed-form z^* , but they do not enforce structural contraction. Our Contractive Linearizer is a general-purpose architecture without inductive bias for grid topology; as noted in §4.1, VIN’s advantage on grid tasks is precisely this architectural match. Crucially, VIN’s operator diverges under repeated application in our experiments, confirming that differentiable VI does not imply contracted VI.

Lipschitz constraints and equilibrium architectures. Lipschitz-bounded layers [Miyato et al., 2018, Anil et al., 2019, Wang and Manchester, 2023] and spectral-norm constraints in deep RL [Bjorck et al., 2021] stabilize training by bounding weight matrices. Deep equilibrium models [Bai et al., 2019] define a layer’s output as the fixed point of a learned implicit operator; Winston and Kolter [2020] enforce equilibrium uniqueness via monotone-operator theory. We combine both ideas: the Contractive Linearizer is a DEQ whose contraction is simple enough ($\alpha \tanh$ diagonal) to admit a *closed-form* fixed point, and the spectral constraint acts on the *composed* operator rather than on individual weight matrices. We are unaware of prior work combining a DEQ-style equilibrium layer with a structural spectral constraint for value-function parameterization.

361 **Conclusion.** The Contractive Linearizer parameterizes V (or Q) as the closed-form fixed point of a
 362 per-state affine contraction, giving a hard convergence guarantee, $\mathcal{O}(1)$ inference ($43\times$ speedup), and
 363 structurally bounded decoded values (Cor. 1) at minimal expressiveness cost (Prop. 1). Instantiated
 364 for planning, DQN, and PPO, it eliminates Q-value divergence, reduces critic variance, and improves
 365 return; running Atari/MuJoCo experiments will test scaling.

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A Reproducibility and implementation details

This appendix documents every detail required to reproduce the experiments of §4: network architectures, optimizer settings, full hyperparameters (per environment), random-seed protocol, environment versions, hardware, and approximate compute budgets. An anonymized code archive accompanies the submission.

A.1 Hardware and software

All classic-control experiments (CartPole, Acrobot, Pendulum, planning) run on a single CPU. The planning training, Atari DQN runs, and MuJoCo PPO runs use one NVIDIA Quadro GV100 GPU (32 GB). Software stack: PyTorch (1.13+), Gym/Gymnasium, and MuJoCo ≥ 2.3 (for -v4 environments). Atari preprocessing follows the standard wrappers of Mnih et al. [2015].

A.2 Contractive Linearizer module

The shared diagonal contractive operator (used in all three settings) is implemented as follows. Given a state-conditioned context vector $c(s) \in \mathbb{R}^{d_c}$ (the output of the encoder MLP backbone), two heads produce

$$\lambda(s) = \alpha \cdot \tanh(\text{MLP}_\Lambda(c(s))) \in \mathbb{R}^d, \quad b(s) = \text{MLP}_b(c(s)) \in \mathbb{R}^d.$$

The closed-form fixed point uses element-wise division $z^*(s) = b(s) \oslash (\mathbf{1} - \lambda(s))^+$, where $(\cdot)^+ := \max(\cdot, \epsilon)$ with $\epsilon = 10^{-2}$. Both head MLPs have two hidden layers of width 128 with ReLU activations. The final linear layer of MLP_Λ is initialized $\mathcal{U}(-0.01, 0.01)$ with zero bias so that $\lambda(s) \approx 0$ at initialization (i.e. $z^*(s) \approx b(s)$, an identity-like start).

A.3 Planning experiment (§4.1)

Environment. 10×10 grid worlds with stochastic obstacles (density 0.2). Reward: -1 per step, 0 at the absorbing goal. Discount: $\gamma = 0.99$. Maps are randomly generated from a pool of 5,000 pre-generated layouts; the target V^* is computed by exact value iteration.

Architectures. *Map encoder:* two convolutional layers $3 \rightarrow 16 \rightarrow 32$ with ReLU, adaptive average pooling to 4×4 , followed by a fully connected projection to context dimension $d_c = 128$. *Latent dimension:* $d = 64$. *Λ head:* two-layer MLP $128 \rightarrow 128 \rightarrow 64$, ReLU. *b head:* two-layer MLP $128 \rightarrow 256 \rightarrow 100$, ReLU. *Decoder/head h :* linear $64 \rightarrow 100$ (one value per grid cell).

Optimization. Adam, learning rate 3×10^{-4} , cosine annealing. Batch size 32, 8,000 training steps, gradient clipping at max-norm 5.0. Loss: $\mathcal{L}_{\text{Bellman}} + 0.5 \mathcal{L}_{\text{fp}} + 0.1 \mathcal{L}_{\text{multi}}$ with $K_{\text{multi}} = 5$.

Baselines. *Unconstrained:* same architecture as Contractive but with no spectral clamp on Λ (Λ is the unconstrained output of the same MLP). *Iterative MLP:* a 3-layer MLP rolled out $K = 5$ times; same parameter count. *VIN:* reproduction of Tamar et al. [2016] with $K = 30$ iterations and 10×10 value/Q maps.

Compute. Single-CPU training takes ≈ 25 min per architecture; total planning-experiment budget < 4 CPU-hours.

A.4 DQN experiments (§4.2)

Classic control (CartPole-v1, Acrobot-v1). Backbone: 2-layer MLP $|\mathcal{S}| \rightarrow 128 \rightarrow 64$ with ReLU. For CONTRACTIVE DQN, the final layer is replaced by the Linearizer head with context $d_c = 64$, latent $d = 32$, and per-action linear heads $\text{head}_a : \mathbb{R}^{32} \rightarrow \mathbb{R}$. Replay buffer 10^4 , batch size 64, discount $\gamma = 0.99$, target-network hard update every 500 steps, ϵ linearly decayed from $1.0 \rightarrow 0.01$ over the first 5,000 (CartPole) / 10,000 (Acrobot) steps. Learning rate: 10^{-3} on CartPole, 5×10^{-4} on Acrobot. Training horizon: 50,000 steps (CartPole), 100,000 steps (Acrobot). Five random seeds.

500 **Atari (Pong, Breakout).** Standard preprocessing [Mnih et al., 2015]: grayscale 84×84 , 4-frame
 501 stack, frameskip 4, reward clipping $[-1, 1]$, no-op reset. Backbone: Nature CNN ($\text{Conv}(4, 32, k =$
 502 $8, s = 4) \rightarrow \text{Conv}(32, 64, k = 4, s = 2) \rightarrow \text{Conv}(64, 64, k = 3, s = 1) \rightarrow \text{FC}(512)$, ReLU).
 503 CONTRACTIVE DQN replaces the final FC with the Linearizer head ($d_c = 512, d = 256$). Replay
 504 buffer 10^5 , batch size 64, $\gamma = 0.99$, target hard update every 1,000 steps, $\epsilon : 1.0 \rightarrow 0.01$ over
 500,000 steps, learning rate 10^{-4} . Training horizon: 2×10^6 env steps, three random seeds.

505 **Stress tests.** *No-target / high-LR*: target network disabled and learning rate $10 \times$ the default (10^{-2}
 506 on CartPole) to provoke divergence. *Poisoned replay*: 20% of replay entries replaced by $(s, a, s', -r)$
 507 (sign-flipped reward) at random times, 5 seeds.

508 **Compute.** CartPole/Acrobot: ≈ 5 CPU-min per seed. Atari: ≈ 8 GPU-hours per seed (single
 509 GV100).

510 A.5 PPO experiments (§4.3)

511 **Architectures.** *Actor (discrete)*: 2-layer MLP $|\mathcal{S}| \rightarrow 64 \rightarrow 64 \rightarrow |\mathcal{A}|$ with tanh, output logits. *Actor*
 512 *(continuous)*: same trunk, with output mean and a learnable log-std parameter (independent of state).
 513 *Standard critic*: 2-layer MLP $|\mathcal{S}| \rightarrow 64 \rightarrow 64 \rightarrow 1$, tanh. *Contractive critic*: two parallel branches
 514 with the same trunk $|\mathcal{S}| \rightarrow 64 \rightarrow 64$, producing scalar $\lambda(s), b(s) \in \mathbb{R}$ (latent dim $d = 1$); the value
 515 is read off in closed form $V(s) = b(s)/(1 - \lambda(s))^+$. Only the critic differs between Standard and
 516 Contractive PPO.

517 **Hyperparameters.** PPO clip 0.2; GAE $\lambda_{\text{GAE}} = 0.95$; discount $\gamma = 0.99$; mini-batch 64; gradient
 518 clip 0.5; value-loss coefficient 0.5; entropy coefficient 0.01 (CartPole) and 0 (others). Rollout length
 519 and PPO epochs: 512/4 for classic control (CartPole, Acrobot, Pendulum); 2,048/10 for MuJoCo.
 520 Learning rate: 3×10^{-4} for both actor and critic.

521 **Environments and horizons.** CartPole-v1 (2×10^5 steps), Acrobot-v1 (5×10^5), Pendulum-v0
 522 (3×10^5), HalfCheetah-v4, Hopper-v4, Walker2d-v4 (10^6 each). Three seeds per environment.

523 **Compute.** Classic-control PPO: ≈ 10 CPU-min per seed. MuJoCo: ≈ 4 GPU-hours per seed
 524 (single GV100).

525 A.6 Random seeds and statistical reporting

526 Seeds are integers $\{0, 1, 2, 3, 4\}$ for 5-seed runs and $\{0, 1, 2\}$ for 3-seed runs. Each seed determinis-
 527 tically initializes Python’s random, NumPy, and PyTorch RNGs as well as the Gym environment.
 528 Tables report mean $\pm$ sample standard deviation; figures shade $\pm 1\sigma$ around the mean curve. The
 529 experiments reported in this paper required no preliminary runs beyond the spectral-bound ablation
 530 in Exp E (§4.1). Total project compute: ~ 50 GPU-hours and ~ 30 CPU-hours.

531 B Structural ablations (Exp F, Exp G)

532 This appendix records the planning-task ablations referenced in §4.1. Both runs use the same 10×10
 533 stochastic-obstacle generator, training schedule, and seeds as Exp A; the only changes are to the
 534 contractive head. *Exp F* ($\lambda \equiv 0$) clamps the eigenvalue MLP’s output to zero throughout training,
 535 leaving $z^*(s) = b(s)$. *Exp G* (*dense* Λ) replaces the diagonal head with a matrix-output head, projects
 536 $\Lambda(s)$ to spectral radius $< \alpha$ by SVD clipping ($\Lambda \rightarrow U \min(\Sigma, \alpha) V^\top$), and computes $z^*(s)$ via the
 537 closed-form linear solve $(\mathbf{I} - \Lambda(s))z^* = b(s)$ at $\mathcal{O}(d^3)$ cost per state. A plain MLP baseline (no
 538 Linearizer head) is included for reference.

539 C Additional notes

540 The *Iterative MLP* baseline in §4.1 unrolls a fixed 3-layer MLP for $K = 5$ steps, with the same
 541 parameter budget as the Contractive model. The *Unconstrained* baseline is identical to Contractive
 542 except that $\Lambda(s)$ is the raw output of MLP_Λ with no α tanh clamp; this is the only architectural
 543 difference, isolating the role of the spectral constraint.

Table 6: Structural ablations on the planning task. Same setup as Exp A, 3 seeds, mean $\pm$ std.

Variant	RMSE @ $K = 50$ ($\downarrow$)	Inference cost
Contractive (diagonal, ours)	<i>pending</i>	$\mathcal{O}(d)$
Dense Λ (Exp G)	<i>pending</i>	$\mathcal{O}(d^3)$
$\lambda \equiv 0$ (Exp F)	<i>pending</i>	$\mathcal{O}(d)$
Plain MLP (no Linearizer)	<i>pending</i>	$\mathcal{O}(d)$

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The four claims listed in the abstract and §1 (a contractive learned operator, $43\times$ closed-form speedup, DQN stability and benchmark gains, PPO critic-variance reduction) are each substantiated by §3 (theory) and §4 (experiments). MuJoCo PPO and Atari DQN results are still running and are explicitly marked as pending.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

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